NCES 2026 • ENVIRONMENTAL SOLUTIONS

Do Microbes Have a Memory?

History-dependent behaviour in changing environments: what published research shows

Vasani Mitul Jinabhai

Department of Life Sciences & Microbiology, Bhakta Kavi Narsinh Mehta University (BKNMU), Junagadh, Gujarat, India

1 The Big Idea

Cells keep meeting changing environments. A past exposure can make the response faster or stronger when the same condition returns, even in daughter cells that never met the first cue.

Main question: does a microbe's past experience change how it responds later?

Not human-style memory. The review calls it history-dependent behaviour (HDB): a response based on epigenetic memory of earlier exposure.

1st exposure → Gap / recovery → Same condition returns → Faster response?

2 Background

- Keeping sensing systems active costs fitness, and sudden changes may leave no time to react.
- Yeast cope better with severe stress after a recent mild stress (priming).
- The lag phase gets shorter when cells have met a carbon source before.
- In bacteria, E. coli prepares for low oxygen when temperature rises.

3 Aim & Source

Aim: to summarise what published research shows about history-dependent (memory-like) responses of microbes in changing environments.

Source reviewed: Vermeersch et al. (2022), Frontiers in Microbiology 13:1004488, an open-access review (CC BY).

- Identify mechanisms behind HDB
- Collect real examples and numbers
- Note costs, trade-offs and limits

4 Three Mechanisms

Chromatin state
Semi-stable changes at gene promoters last a few cell generations, so genes switch on faster next time.

Protein inheritance
Proteins made at the first exposure persist and pass to daughter cells, e.g. protective proteins after high salt.

Metabolic state
Respiratory activity, mitochondria and ATP decide how fast cells restart growth after a shift.

5 Case Study: Sugar Shift

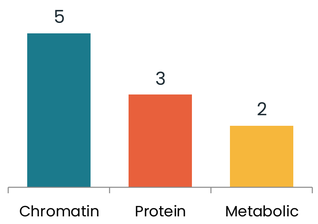
1. Yeast grow on glucose, then shift to a less preferred sugar (maltose or galactose).
2. ATP drops and growth stops. This pause is the lag phase.
3. Respiration genes switch on first. Only cells that restore ATP resume growth.
4. A second shift soon after the first is often smoother, with a shorter lag.
5. Longer time on glucose means more repressed respiration and a longer lag.

Single yeast cells in one population: some restart within ~5 h, others need >20 h, and some never restart.

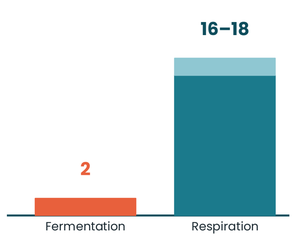
6 Graphic Data Analysis

All numbers are taken from the reviewed paper. Add your own graphs and images in the dashed boxes.

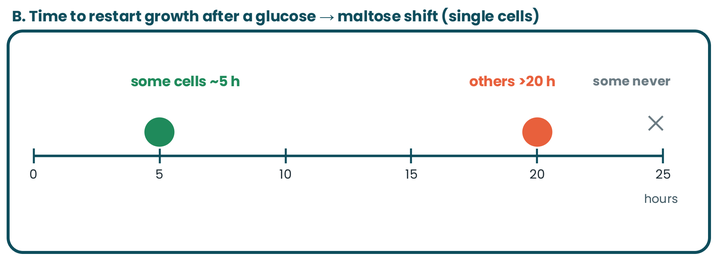
A. Example conditions listed per mechanism



C. ATP made per glucose molecule



B. Time to restart growth after a glucose → maltose shift (single cells)



7 Biostatistics

Mechanism	Conditions listed	Share
Chromatin state	5	50%
Protein inheritance	3	30%
Metabolic state	2	20%
Total	10 entries (7 distinct)	100%

Descriptive tally from Figure 2 of the review. Carbon-source shift appears under all 3 mechanisms.

No p-value or meta-analysis here: the source is a narrative review without pooled numerical data. Next step: extract n, mean and 95% CI from original papers; combine only if studies are comparable.

8 Principle of HDB

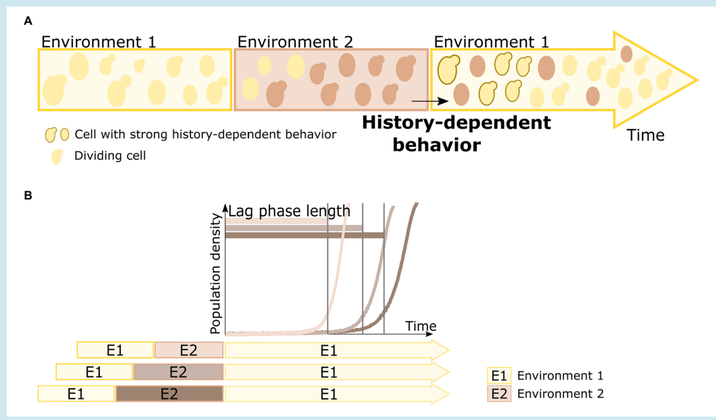


Figure 1 (A, B) from Vermeersch et al. 2022, CC BY 4.0. The longer cells stay in environment 2, the longer the lag when environment 1 returns.

9 Mobile-Size Box

Use this box for

- QR code to the paper (DOI)
- A mini infographic
- Your own photo or short-video preview

10 Key Findings

- Past exposure can speed up a later response (shorter lag).
- Chromatin, inherited proteins and metabolic state all contribute.
- Respiration and ATP recovery decide which cells restart growth.
- Cells in one population can respond very differently.

11 Costs & Trade-offs

- Strong HDB can cost growth: yeast evolved in glucose/maltose cycles had shorter lags but grew slower on glucose.
- Early gene induction gives a shorter lag later, but slower growth now.
- Mixed responses within a population may act as bet-hedging.

12 Limitations & Open Questions

- Almost all studies are lab-based; importance in nature is largely unknown.
- Methods and strains differ, so comparisons are hard.
- Most evidence is from yeast; fewer bacterial examples.
- Link to pollution is untested here (open question).

Reference: Vermeersch L, Cool L, Gorkovskiy A, Voordeckers K, Wenseleers T, Verstrepen KJ (2022). Do microbes have a memory? History-dependent behavior in the adaptation to variable environments. Front. Microbiol. 13:1004488. doi: 10.3389/fmicb.2022.1004488 (CC BY 4.0). Figure 1 reproduced under CC BY.
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